

	Describe in details GSM architecture with necessary block diagram and its various blocks	
	What is hand over in GSM? Give comparison of hard hand over and soft hand over	
	Compare CDMA forward and reverse channel.	
	Define the following terms regarding wireless communication: (1) Control channel (2) Half duplex channel (3) Base station (4) Mobile Switching Center (5) Simplex systems (6) Full duplex systems (7) Reverse channel	
	(1) With figure explain the concept of frequency reuse in detail. (2) With figure explain hand off scenario at cell boundary	
	Explain the concept of Cell splitting in detail with figure	
	Explain the concept of sectoring in detail with figure	
	Explain the concept of Microcell Zoning.	
	If GSM uses a frame structure where each frame consists of 8 time slots and each time slot contains 156.25 bits and data is transmitted at 270.833 kbps in the channel find:(i) The time duration of a bit. (ii) The time duration of a slot. (iii) The time duration of a frame.(iv) How long must a user occupying a single time slot wait between two successive transmissions	
	If a normal GSM time slot consists of 6 trailing bits, 8.25 guard bits, 26 training bits and two traffic bursts of 58 bits of data. Find the frame efficiency.	
	Briefly explain GSM logical channels.	
	Determine frame efficiency of a TDMA frame structure used in GSM system.	
	Explain Machine Type Communications (MTC) in 5G. Discuss mMTC, URLLC, and their applications in IoT.	
	Draw and explain concept of TDMA frames,full-rate and half-rate traffic channels, and the control channels	
	Draw and explain the architecture of UMTS with its working principle, How did LTE evolved from UMTS?	
	If a signal to interference ratio of 15 dB is required for satisfactory forward channel performance of a cellular system, what is the frequency reuse factor and cluster size that should be used for maximum capacity if the path loss exponent is (1) $n=4$ (2) $n=3$?Assume that there are six co channel cells in the first tier and all of them are at the same distance from the mobile. Use suitable approximations.	
	How co-channel interference and system capacity are related?	
	Define and explain mean excess delay, rms delay spread and excess delay spread.	
	With the help of timing parameters, explain frame structure for GSM.	
	How does GSM radio subsystem work?	
	Give differences between HLR and VLR functions.	
	A cellular system has 32 cells; each cell has 1.6 km radius and the system reuse factor of 7. The system is to support 336 traffic channels in total. Determine the	

	total geographical area covered, the number of traffic channels per cell and total number of simultaneous calls supported by this system.	
	For a regular hexagonal geometry show that co-channel reuse ratio is $Q = (3N)^{1/2}$, where $N = i^2 + j^2$.	
	If 20 MHz of total spectrum is allocated for a duplex wireless cellular system and each simplex channel has 25 kHz RF bandwidth, find:(i) The number of duplex channels(ii) The total number of channels per cell site ,if N=4 cell reuse is used	7
	Compare the S/I ratio for a mobile radio for the following cases of cellular system with frequency reuse factor of 7 for (i) Omni directional antenna (ii) 120° sectoring (iii) 60° Sectoring.	
	Define following terms: Cell, Cluster, Cluster Size, Voice Channel, Control Channel, Forward Channel, Reverse Channel, Channel Capacity, Adjacent Channel Interference, Co-channel Interference, Co-channel Reuse Ratio, Frequency Reuse (Frequency Planning), Frequency Reuse Factor, Footprint, Simplex system, Full Duplex System, Paging System, Page, Base Station (BS), Mobile Switching Center (MSC), Handoff, Inter System handoff (Inter MSC Handoff), MCHO, NCHO, MAHO, Umbrella Cell, Cell Dragging, Breathing Cell	
	What is mMTC in 5G? How does it work?	
	Massive Machine-Type Communications (mMTC) is one of three core 5G service areas. It has been created specifically to enable a huge volume of small data packets to be collected from large numbers of devices, simultaneously. mMTC supports applications using Internet of Things (IoT) sensors, meaning that data can be used to reduce energy consumption, make work more efficient, or improve our lives in other ways. The 5G network architecture has been designed with three key service areas in mind: 1. Massive Machine-Type Communications (mMTC). This is used to connect large numbers of devices and is expected to transform the IoT industry. It aims to meet the demands of smart cities and other IoT applications. 2. Enhanced Mobile Broadband (eMBB). This service area is essentially an extension and improvement on that which is currently used for cellular data. It provides speeds of up to 10Gbps for high bandwidth usage such as HD video streaming or VR/AR gaming, providing faster download speeds and improved user experiences.	

	3. Ultra-reliable Low-Latency Communication (URLLC). URLLC is ideal for mission-critical applications where network reliability and low-latency (less than 1 ms) is paramount. This could be used in applications such as self-driving cars or industrial remote control. mMTC supports connection densities of up to one million devices per square kilometre. This is over ten times higher than 4G LTE network capability. With this ability, 5G can provide the infrastructure needed to support huge networks of cellular-connected sensors. The mMTC service area is designed for massive IoT deployments using large numbers of low-power devices to regularly transmit small amounts of data. IoT sensors transmit small data packets over a relatively long distance (up to 10 km). This makes them perfectly placed for applications such as smart agriculture; where hundreds of sensors send weather, soil moisture and fill-level data across areas of remote farmland. Using this data, farmers can optimise irrigation frequency and volumes to ensure high crop yields. But without the low latency networks to connect these sensors in rural, deep indoors or underground locations, they would fail to function. Network slicing, also called software-defined networking or SDN technology, is the system of dividing bandwidth up across these different areas, allowing different service functions to be customised. This means each application can access the required resource simultaneously, with higher priority applications prioritised. mMTC offers low-bandwidth connectivity with deep coverage, whereas URLLC provides high reliability without the same depth of coverage. Existing IoT technology uses low-power wide area networks (LPWAN), which often require an ethernet or cellular backhaul from an IoT gateway. The sensors operate on a network such as LoRaWAN or Sigfox, then the gateway sends the information to a server using a 4G LTE network or a wired internet connection. mMTC offers exciting opportunities to acquire and exploit large datasets, however meeting design specifications for small, low-cost, IoT devices can prove challenging. Particularly when the long-range and low power consumption requirements are considered, placing specific requirements and constraints on designers.	
	Draw the block diagram of MIMO system model. Derive the expression of MIMO channel capacity.	7
	What is Massive MIMO? Discuss the advantages and challenges of MASSIVE MIMO for 5th generation mobile technology.	7
	Compare Direct Sequence Spread Spectrum (DSSS) and Frequency Hopping Spread Spectrum (FHSS) relevant to GSM and IS-95.	
	Explain the near-far problem and how IS-95 solves it.	
	Explain LTE Release-10 enhancements over LTE Release-8/9. Include carrier aggregation, enhanced MIMO, relay nodes, and heterogeneous networks.	
	Explain mmWave communications. Discuss challenges (path loss, blockage, penetration loss) and techniques (beamforming, hybrid beamforming, large arrays)	

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